Chapter 5: Central-Force Motion

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In this chapter, we will consider a seemingly abstract problem: the motion of a particle under the influence of a force that is always directed towards or away from a fixed point in space. This type of force is called a *central force*. Although this may initially seem like an abstract problem, since in reality the body exerting the force is not necessarily fixed in space, it is of great importance in physics. In particular, it can be used to model the motion of planets and satellites under the influence of gravity, the motion of electrons around atomic nuclei under the influence of electrostatic forces, and many other physical phenomena.

We will begin by defining what a central force is, and then we will derive some important consequences of this definition. We will redefine velocity and acceleration in polar coordinates, which are more suitable for describing central-force motion. We will then derive the equations of motion for a particle under the influence of a central force, and we will discuss how to solve these equations to determine the trajectory of the particle.

1 Definition of Central Force

A force is central if:

(i) it is towards or away from a fixed point called *the center of force*.

(ii) the magnitude of the force is only proportional to the distance from the center of force.

Above assumptions can mathematically be expressed as

$$\mathbf{F}(\mathbf{r}) = f(r)\mathbf{e}_r, \tag{1}$$

where $f(r)$ is a scalar function of the distance r from the center of force, and $\mathbf{e}_r$ is the unit vector in the direction of the position vector $\mathbf{r}$.

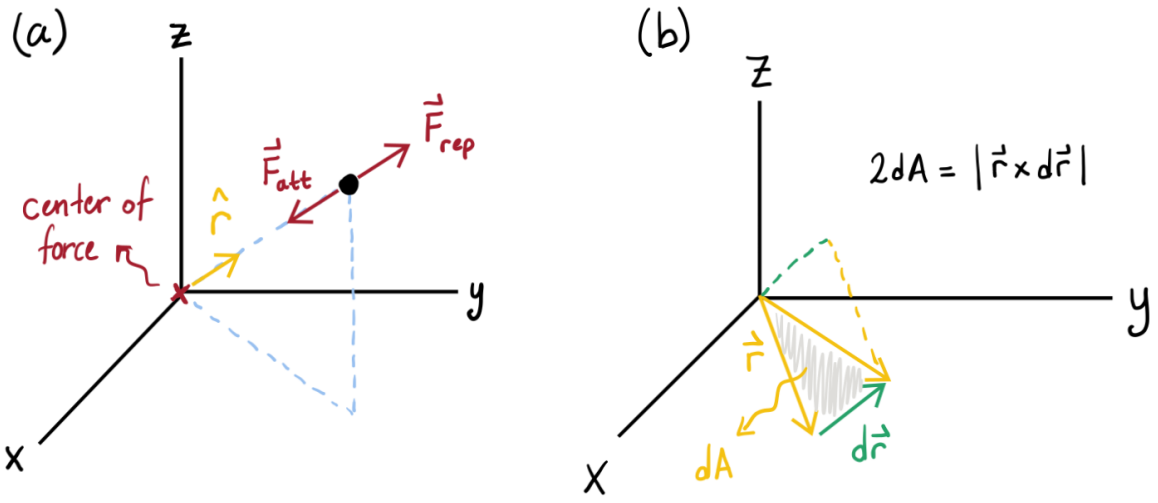


Figure 1: A central force is always directed towards or away from a fixed point in space, called the center of force. The magnitude of the force depends only on the distance from the center of force.

2 Direct Consequences of the Definitions

From Eq. 1, we can draw 3 direct important conclusions:

(i) Central force cannot apply any torque on the particle. Therefore, the angular momentum of the particle is conserved.

$$\tau = \mathbf{r} \times \mathbf{F} = \mathbf{r} \times (f(r)\mathbf{e}_r) = f(r)\mathbf{r} \times \mathbf{e}_r = 0.$$

since $\mathbf{r}$ and $\mathbf{e}_r$ are parallel. Therefore, the angular momentum of the particle is conserved:

$$\tau = \frac{d\mathbf{L}}{dt} = 0 \implies \mathbf{L} = \text{constant}.$$

(ii) The motion of the particle is confined to a plane. This is because the angular momentum vector $\mathbf{L} = \mathbf{r} \times \mathbf{p}$ is constant in direction, and the position vector $\mathbf{r}$ always lies in the plane perpendicular to $\mathbf{L}$.

(iii) The position vector of the particle sweeps out equal areas in equal times, i.e., the areal velocity is constant. This is a direct consequence of the conservation of angular momentum. The areal velocity is given by:

$$dA = \frac{1}{2}|\mathbf{r} \times d\mathbf{r}| = \frac{1}{2}|\mathbf{r} \times \mathbf{v}dt| \implies \frac{dA}{dt} = \frac{1}{2}|\mathbf{r} \times \mathbf{v}| = \frac{1}{2m}|\mathbf{r} \times m\mathbf{v}| = \frac{L}{2m},$$

where L is the magnitude of the angular momentum and m is the mass of the particle. Since L is constant, the areal velocity dA/dt is also constant, i.e, $A(t)$ is a linear function of time.

3 Polar Coordinates

We have been using Cartesian coordinates to describe the motion of a particle. However, Cartesian coordinates are not the best choice for central-force motion as motion involves forces whose directions change with the position of the particle.

The central force always acts along the line connecting the particle to a fixed point, which we choose as the origin. Furthermore, since the force produces no torque about this point, the angular momentum of the particle is conserved, and its motion is confined to a plane. It is therefore more convenient to describe the motion using polar coordinates, in which the position of the particle is specified by its distance r from the origin and the angle θ it makes with a fixed reference direction as shown in Fig. 2(a).

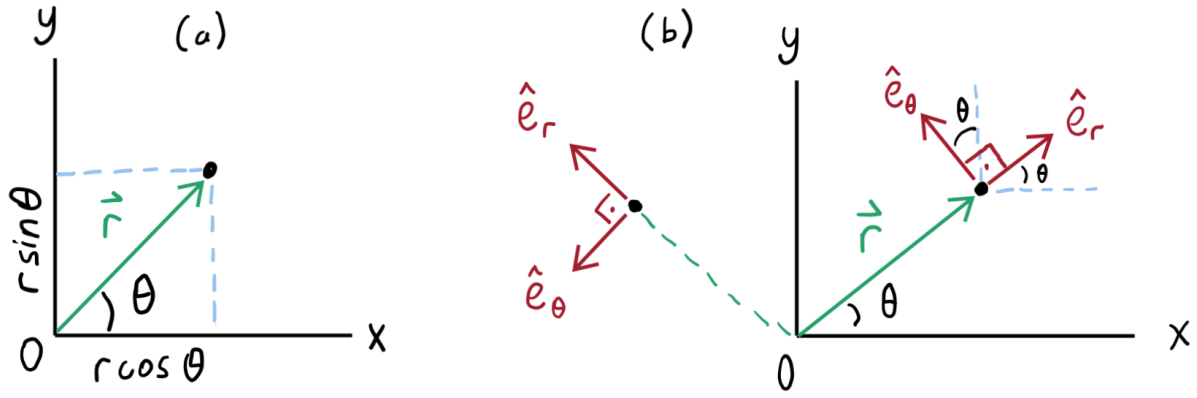


Figure 2: (a) The position of a particle in polar coordinates is specified by its distance r from the origin and the angle θ it makes with a fixed reference direction. (b) The unit vectors $\mathbf{e}_r$ and $\mathbf{e}_\theta$ are not fixed in direction, but change as the particle moves.

The cartesian coordinates (x, y) of the particle are related to its polar coordinates (r, θ) by the equations:

$$x = r\cos\theta, \quad y = r\sin\theta.$$

The main advantage of polar coordinates is that one of the coordinate directions is always aligned with the central force. This allows us to separate the radial and angular components of the motion, considerably simplifying the equations of motion.

It is worth noting that we still want to locate the particle in a two-dimensional plane. However, we change our definition for locating the particle: Instead of measuring the perpendicular distance from two perpendicular axes as in Cartesian coordinates, we measure the distance from a single point (the origin) and the angle from a fixed direction. This simple change in perspective will allow us to write down the velociy and acceleration in the following section to contain some new information about the motion of the particle which do not show up in Cartesian coordinates.

In polar coordinates, unit vectors are not fixed in direction. Their directions change as the particle moves as shown in Fig. 2(b). The unit vector $\mathbf{e}_r$ points radially outward from the

origin, while the unit vector $\mathbf{e}_\theta$ is perpendicular to $\mathbf{e}_r$ and points in the direction of increasing θ. The position vector of the particle can be expressed as:

$$\mathbf{r} = r\mathbf{e}_r.$$

We can write polar unit vectors in terms of Cartesian unit vectors as:

$$\mathbf{e}_r = \cos\theta\mathbf{i} + \sin\theta\mathbf{j}, \quad \mathbf{e}_\theta = -\sin\theta\mathbf{i} + \cos\theta\mathbf{j}.$$

Time derivatives of the polar unit vectors are:

$$\frac{d\mathbf{e}_r}{dt} = -\dot{\theta}\sin\theta\mathbf{i} + \dot{\theta}\cos\theta\mathbf{j} = \dot{\theta}\mathbf{e}_\theta, \quad \frac{d\mathbf{e}_\theta}{dt} = -\dot{\theta}\cos\theta\mathbf{i} - \dot{\theta}\sin\theta\mathbf{j} = -\dot{\theta}\mathbf{e}_r.$$

Or, in a more compact form:

$$\dot{\mathbf{e}}_r = \dot{\theta}\mathbf{e}_\theta, \quad \dot{\mathbf{e}}_\theta = -\dot{\theta}\mathbf{e}_r. \tag{2}$$

Geometrically, these equations can be interpreted as follows: As the particle moves, the unit vectors, $\mathbf{e}_r$ and $\mathbf{e}_\theta$, cannot increase in magnitude. The only way for them to change is through rotation. The unit vector $\mathbf{e}_r$ rotates in the direction of $\mathbf{e}_\theta$ with an angular speed of $\dot{\theta}$: as we increase the angle θ, the unit vector $\mathbf{e}_r$ automatically rotates in the direction $\mathbf{e}_\theta$. Similarly, as θ increases, the unit vector $\mathbf{e}_\theta$ rotates in the direction opposite to $\mathbf{e}_r$ with an angular speed of $\dot{\theta}$.

The infinitesimal displacements in two perpendicular directions in polar corrdinates are given as dr and $r\,d\theta$ as shown in Figure 3. Therefore, we can write the infinitesimal displacement as:

$$d\mathbf{r} = dr\,\mathbf{e}_r + r\,d\theta\,\mathbf{e}_\theta.$$

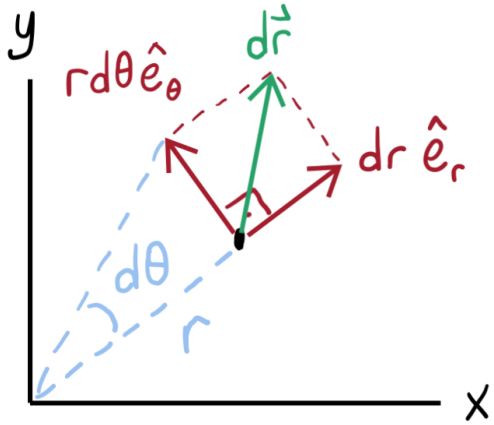


Figure 3: The infinitesimal displacement of a particle in polar coordinates can be expressed as the sum of two perpendicular displacements: dr in the radial direction and $r\,d\theta$ in the angular direction.

4 Velocity, Acceleration and Angular Momentum in Polar Coordinates

The velocity of the particle is the time derivative of its position vector:

$$\mathbf{v} = \dot{\mathbf{r}} = \frac{d}{dt}(r\mathbf{e}_r) = \dot{r}\mathbf{e}_r + r\dot{\mathbf{e}}_r.$$

Using Eq. 2, we can write

$$\mathbf{v} = \dot{r}\mathbf{e}_r + r\dot{\theta}\mathbf{e}_\theta.$$

If we denote the radial and angular components of the velocity as v_r and v_θ, respectively, we have:

$$v_r \coloneqq \dot{r}, \quad v_\theta \coloneqq r\dot{\theta} \implies \mathbf{v} = v_r\mathbf{e}_r + v_\theta\mathbf{e}_\theta.$$

Geometrically, the radial component of the velocity, v_r, represents the rate at which the distance from the origin is changing, while the angular component, v_θ, represents the rate at which the angle θ is changing. The angular component can also be interpreted as the tangential speed of the particle along a circular path of radius r: $\frac{rd\theta}{dt} = \frac{ds}{dt}$. The speed of the particle is given by the magnitude of the velocity vector:

$$v = |\mathbf{v}| = \sqrt{v_r^2 + v_\theta^2} = \sqrt{\dot{r}^2 + (r\dot{\theta})^2}.$$

The acceleration of the particle is the time derivative of its velocity vector:

$$\mathbf{a} = \dot{\mathbf{v}} = \frac{d}{dt}(\dot{r}\mathbf{e}_r + r\dot{\theta}\mathbf{e}_\theta) = \ddot{r}\mathbf{e}_r + \dot{r}\dot{\mathbf{e}}_r + \dot{r}\dot{\theta}\mathbf{e}_\theta + r\ddot{\theta}\mathbf{e}_\theta + r\dot{\theta}\dot{\mathbf{e}}_\theta.$$

Using Eq. 2, we can write

$$\mathbf{a} = (\ddot{r} - r\dot{\theta}^2)\mathbf{e}_r + (r\ddot{\theta} + 2\dot{r}\dot{\theta})\mathbf{e}_\theta. \tag{3}$$

The first term in the radial component of the acceleration, $\ddot{r}$, represents the rate of change of the radial velocity, while the second term, $-r\dot{\theta}^2 = -r\omega^2$, represents the centripetal acceleration that arises due to the change in direction of the velocity vector as the particle moves along a curved path. The first term in the angular component of the acceleration, $r\ddot{\theta}$, represents the rate of change of the angular velocity, while the second term, $2\dot{r}\dot{\theta}$, represents the *Coriolis acceleration* that arises due to the change in direction of the radial velocity as the particle moves along a curved path. The Coriolis accelaration will be discussed in more detail in Chapter 7: Motion in Non-Inertial Reference Frames.

If we denote the radial and angular components of the acceleration as a_r and a_θ, respectively, we can write the acceleration in a compact form as:

$$a_r \coloneqq \ddot{r} - r\dot{\theta}^2, \quad a_\theta \coloneqq r\ddot{\theta} + 2\dot{r}\dot{\theta} \implies \mathbf{a} = a_r\mathbf{e}_r + a_\theta\mathbf{e}_\theta.$$

Following its definition, we can write the angular momentum of the particle in polar coordinates as:

$$\mathbf{L} = \mathbf{r} \times m\mathbf{v} = (r\mathbf{e}_r) \times m(\dot{r}\mathbf{e}_r + r\dot{\theta}\mathbf{e}_\theta) = mr^2\dot{\theta}(\mathbf{e}_r \times \mathbf{e}_\theta).$$

Since $\mathbf{e}_r \times \mathbf{e}_\theta$ is a unit vector perpendicular to the plane of motion, we can write

$$\mathbf{L} = mr^2\dot{\theta}\mathbf{k},$$

where $\mathbf{k}$ is a unit vector perpendicular to the plane of motion. Note that even if the quantities r and $\dot{\theta}$ change with time, the product

$$L = mr^2\dot{\theta} \tag{4}$$

remains constant due to the conservation of angular momentum.

5 Equation of Motion for Central-Force Motion

Plugging the expressions for the central force and acceleration in polar coordinates, as given in Eq. 1 and Eq. 3, into Newton's second law,

$$f(r)\mathbf{e}_r = m(\ddot{r} - r\dot{\theta}^2)\mathbf{e}_r + m(r\ddot{\theta} + 2\dot{r}\dot{\theta})\mathbf{e}_\theta.$$

and equating the coefficients of $\mathbf{e}_r$ and $\mathbf{e}_\theta$ on both sides, we obtain the equations of motion for a particle under the influence of a central force:

$$m(\ddot{r} - r\dot{\theta}^2) = f(r), \quad m(r\ddot{\theta} + 2\dot{r}\dot{\theta}) = 0. \tag{5}$$

Here, the first equation describes the radial motion of the particle, while the second equation describes the angular motion. Since we know that $a_\theta = r\ddot{\theta} + 2\dot{r}\dot{\theta}$, the second equation tells us that no net force is acting on the particle in the tangential direction. Multiplying the second equation by r corresponds to a zero torque, and leads to the conservation of angular momentum:

$$mr(r\ddot{\theta} + 2\dot{r}\dot{\theta}) = \frac{d}{dt}(mr^2\dot{\theta}) = \frac{dL}{dt} = 0.$$

Hence, the second equation is a restatement of the conservation of angular momentum. The first equation can be rewritten as:

$$m\ddot{r} = f(r) + mr\dot{\theta}^2 = f(r) + \frac{L^2}{mr^3},$$

where we have used the fact that $L = mr^2\dot{\theta}$ is constant. This equation describes the radial motion of the particle in terms of two contributions: the central force $f(r)$ and an additional term arising from the angular motion. The latter can be interpreted as an *effective centrifugal force*,

$$F_{\text{cent}}(r) = \frac{L^2}{mr^3},$$

which is always directed away from the origin. This term arises from the rotation of the polar coordinate basis and can equivalently be understood as the fictitious centrifugal force experienced in a reference frame rotating with the radial direction. It is not an additional

physical interaction, but rather a consequence of describing the motion using a rotating coordinate basis. We can therefore define the *effective radial force* as

$$f_{\text{eff}}(r) = f(r) + \frac{L^2}{mr^3}, \tag{6}$$

so that the radial equation of motion takes the familiar one-dimensional form

$$m\ddot{r} = f_{\text{eff}}(r).$$

6 Binet's Equation

By using the radial and angular equations of motion in Eq. 5, we can derive a second-order differential equation that relates the trajectory of the particle to the central force acting on it. This equation is known as *Binet's equation*. Starting from the radial equation of motion, we have

$$\ddot{r} - r\dot{\theta}^2 = \frac{f(r)}{m}.$$

We will write the terms in left-hand side in a more convenient form. From Eq. 4, we know $r^2\dot{\theta} := h$, where h is a constant. Therefore, we can write

$$\dot{r} = \frac{dr}{d\theta}\dot{\theta} = \frac{h}{r^2}\frac{dr}{d\theta} \implies \ddot{r} = \frac{h}{r^2}\frac{d}{d\theta}\left(\frac{h}{r^2}\frac{dr}{d\theta}\right)$$

Change of variable

$$u = \frac{1}{r} \implies \frac{du}{d\theta} = -\frac{1}{r^2}\frac{dr}{d\theta} \implies \frac{h}{r^2}\frac{dr}{d\theta} = -h\frac{du}{d\theta} \implies \ddot{r} = -\frac{h^2}{r^2}\frac{d^2u}{d\theta^2}.$$

Also

$$\dot{\theta} = \frac{h}{r^2} = hu^2 \implies r\dot{\theta}^2 = \frac{h^2}{r^3} = h^2u^3.$$

Plugging these expressions into the radial equation of motion, we obtain

$$-\frac{h^2}{r^2}\frac{d^2u}{d\theta^2} - h^2u^3 = \frac{f(r)}{m} \implies \frac{d^2u}{d\theta^2} + u = -\frac{f(1/u)}{mh^2u^2}.$$

7 Mechanical Energy and Equation of Motion

We know that if a given force is conservative and the potential energy of the system does not explicitly depend on time, then the mechanical energy of the system is conserved. The infinitesimal work done by the central force is

$$dW = \mathbf{F}\cdot d\mathbf{r} = f(r)\mathbf{e}_r \cdot (dr\,\mathbf{e}_r + r\,d\theta\,\mathbf{e}_\theta) = f(r)\,dr,$$

since $\mathbf{e}_r \cdot \mathbf{e}_\theta = 0$, see Fig. 4. Therefore, the total work done by the force in moving a particle from point A to point B is

$$W_{AB} = \int_{r_A}^{r_B} f(r)\, dr.$$

Since this integral depends only on the initial and final radial distances, and not on the path taken, the central force is *conservative*. In particular, the work done along any closed path vanishes:

$$\oint \mathbf{F} \cdot d\mathbf{r} = \oint f(r)\, dr = 0.$$

Consequently, we can define a potential energy $U(r)$ such that

$$f(r) = -\frac{dU}{dr}.$$

Thus, every position-dependent central force is conservative wherever the force is well-defined. The mechanical energy of the particle is given by

$$E = K + U(r) = \frac{1}{2}m(\dot{r}^2 + r^2\dot{\theta}^2) + U(r) = \frac{1}{2}m\dot{r}^2 + \frac{1}{2}mr^2\dot{\theta}^2 + U(r),$$

where the first term is the kinetic energy in the radial direction, the second term ($\frac{1}{2}I\omega^2$, with $I = mr^2$) is the kinetic energy in the angular direction, and the third term is the potential energy. Until now, we have only written the mechanical energy in terms of the radial and angular velocities. By plugging in $\dot{\theta} = \frac{L}{mr^2}$, we can incorporate the conservation of angular momentum into the expression and rewrite it as

$$E = \frac{1}{2}m\dot{r}^2 + \frac{L^2}{2mr^2} + U(r) \implies \dot{r} = \pm\sqrt{\frac{2}{m}\left(E - U(r) - \frac{L^2}{2mr^2}\right)}. \tag{7}$$

This equation is the first integral of the radial equation of motion for a particle under a central force. Once $U(r)$ is known, it can be solved for $r(t)$. This equation can also be transformed into another form that is more useful for determining the trajectory of the particle. On an interval where r changes monotonically,

$$d\theta = \frac{d\theta}{dt}\frac{dt}{dr}dr = \frac{L}{mr^2}\frac{dt}{dr}dr = \frac{L}{mr^2}\frac{1}{\dot{r}}dr = \pm\frac{L}{r^2}\frac{dr}{\sqrt{2m(E - U(r)) - \frac{L^2}{r^2}}}.$$

Therefore, the trajectory of the particle can be determined by integrating this equation:

$$\theta(r) = \int \frac{\pm L/r^2}{\sqrt{2m\left[E - U(r) - \frac{L^2}{2mr^2}\right]}}dr. \tag{8}$$

It is not always possible to evaluate this integral analytically, but it can generally be evaluated numerically for a given potential energy function $U(r)$. Some important central-force problems that can be solved analytically are

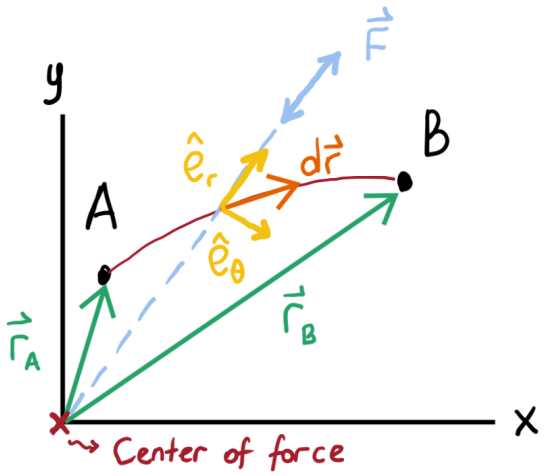


Figure 4: The $\mathbf{e}_\theta$ component of $d\mathbf{r}$ does not contribute to the work done by a central force since it is perpendicular to $\mathbf{F}$.

- isotropic harmonic oscillator $f(r) = -kr$,
- inverse-square law $f(r) = -\frac{k}{r^2}$,
- inverse-cube law $f(r) = -\frac{k}{r^3}$.

The first two are of great importance in physics, while the last one is also useful for studying the properties of central-force motion. We will solve the inverse-square law in detail in Sec. 11.

Note that since L is constant, the time rate of change of θ, $\dot{\theta} = \frac{L}{mr^2}$, cannot change its sign as long as $L \neq 0$ and $r > 0$. Once L is fixed, the particle can only move in one angular direction around the center of force. However, this does not prevent the radial motion from reversing direction, nor does it imply that the trajectory cannot intersect itself. In fact, central-force trajectories can be closed or self-intersecting. For $L = 0$, the motion is purely radial, and the particle may pass through the origin if the force is regular there.

8 Orbits in Central Force Motion

We can define an *effective potential* to simplify the analysis of central-force motion via integrating the effective force as given in Eq. 6:

$$U_{\text{eff}}(r) = -\int f_{\text{eff}}(r)dr = -\int \left(f(r) + \frac{L^2}{mr^3}\right) dr = U(r) + \frac{L^2}{2mr^2} + C, \tag{9}$$

where $U(r)$ is the actual potential energy, $\frac{L^2}{2mr^2}$ is the centrifugal potential energy, and C is a constant of integration. We know that we can always choose the reference of potential

arbitrarily; the value of C does not affect the physical behavior of the system. Therefore, setting $C = 0$, we obtain the effective potential

$$U_{\text{eff}}(r) \coloneqq U(r) + \frac{L^2}{2mr^2},$$

so that the conservation of mechanical energy takes the form

$$\frac{1}{2}m\dot{r}^2 + U_{\text{eff}}(r) = E.$$

Throughout this section, we assume that $L \neq 0$. Since the radial kinetic energy cannot be negative, the particle can access only those radii for which

$$E \geq U_{\text{eff}}(r).$$

Thus, a plot of the effective potential together with the horizontal line representing the total energy provides a useful picture of the radial motion.

8.1 Turning Points and Bounded Motion

At a radial turning point, $\dot{r} = 0$, and hence

$$E = U_{\text{eff}}(r) = U(r) + \frac{L^2}{2mr^2}.$$

An ordinary turning point is a boundary of an allowed radial interval at which the radial motion reverses direction. The form and number of solutions of this equation depend on the potential $U(r)$; it is not generally a quadratic equation in r.

If the allowed radial interval is bounded by two ordinary turning points, $r_{\min}$ and $r_{\max}$, the radial coordinate oscillates between these values, and the orbit is *bounded*. Circular orbits and degenerate turning points require separate consideration. If the allowed radial interval extends to infinity, radially *unbounded* motion is possible.

The relevant interval is the one accessible from the initial conditions: different allowed intervals may be separated by forbidden regions where $E < U_{\text{eff}}(r)$.

Notice that $\dot{r} = 0$ at a turning point does not mean that the particle is at rest. Its tangential speed is

$$v_\theta = r\dot{\theta} = \frac{L}{mr},$$

which is nonzero for $L \neq 0$. Also, bounded motion does not necessarily produce a closed orbit. A *closed orbit* is one in which the particle returns to its initial position after a finite time. The radial coordinate may repeat its motion while the particle returns to a different angular position, producing a *precessing orbit*.

8.2 Circular Orbits and Their Stability

A circular orbit has a constant radius r_0. Therefore, both $\dot{r}$ and $\ddot{r}$ must vanish. Using the radial equation

$$m\ddot{r} = -\frac{dU_{\text{eff}}}{dr},$$

the conditions for a circular orbit are

$$U'_{\text{eff}}(r_0) = 0, \qquad E = U_{\text{eff}}(r_0).$$

Equivalently,

$$U'(r_0) = \frac{L^2}{mr_0^3}.$$

A strict local minimum of the effective potential corresponds to a stable circular orbit, whereas a strict local maximum corresponds to an unstable one. To see this, consider a small radial displacement $r = r_0 + \delta r$ at fixed angular momentum. To first order,

$$m\ddot{\delta r} = -U''_{\text{eff}}(r_0)\delta r.$$

If $U''_{\text{eff}}(r_0) > 0$, the displacement undergoes small oscillations with angular frequency

$$\omega_r = \sqrt{\frac{U''_{\text{eff}}(r_0)}{m}}.$$

If $U''_{\text{eff}}(r_0) < 0$, a radial perturbation can grow, and the circular orbit is unstable. If the second derivative vanishes, higher-order terms must be examined.

9 Effective Potential for Inverse-Square Forces

The inverse-square force is of particular interest because it describes both the gravitational force and the electrostatic force. The potential energy for an attractive inverse-square force is

$$U(r) = -\frac{\alpha}{r}, \qquad \alpha > 0.$$

The corresponding effective potential is

$$U_{\text{eff}}(r) = -\frac{\alpha}{r} + \frac{L^2}{2mr^2}, \tag{10}$$

where L is the conserved angular momentum of the particle about the center of force. The effective potential consists of two competing contributions: the attractive potential energy, which is negative and proportional to $1/r$, and the centrifugal potential energy, which is positive and proportional to $1/r^2$.

For nonzero angular momentum, the centrifugal term dominates as $r \to 0$, causing the effective potential to approach positive infinity. Consequently, the particle cannot reach

the center of force with finite total energy. On the other hand, at large distances, the attractive term dominates, and the effective potential approaches zero from below. The effective potential therefore has a minimum, which can be found by setting its derivative to zero:

$$\frac{dU_{\text{eff}}}{dr} = \frac{\alpha}{r^2} - \frac{L^2}{mr^3} = 0.$$

Solving for r, we obtain

$$r_0 = \frac{L^2}{m\alpha}. \tag{11}$$

At this radius, the effective radial force vanishes, and the particle can move in a circular orbit. The corresponding total energy is

$$E_{\min} = U_{\text{eff}}(r_0) = -\frac{m\alpha^2}{2L^2}. \tag{12}$$

Since the effective potential has a minimum at r_0, the circular orbit is stable against small radial perturbations. The radial motion is determined by conservation of energy:

$$\frac{1}{2}m\dot{r}^2 + U_{\text{eff}}(r) = E. \tag{13}$$

Because the radial kinetic energy cannot be negative, the particle can only occupy regions where

$$E \geq U_{\text{eff}}(r).$$

The turning points of the radial motion satisfy $E = U_{\text{eff}}(r)$. Depending on the total energy, several types of motion are possible, as illustrated in Fig. 5.

1. Energy at the minimum ($E = E_{\min}$): When the total energy equals the minimum of the effective potential, the only possible radial position is $r = r_0$. Since the radial kinetic energy must vanish, we have $\dot{r} = 0$. The particle therefore maintains a constant distance from the center of force and follows a circular orbit.

2. Negative energy ($E_{\min} < E < 0$): For energies above the minimum but below zero, the horizontal energy line intersects the effective potential at two points, $r_{\min}$ and $r_{\max}$. At these turning points, the radial velocity vanishes: $\dot{r} = 0$. The particle can move only within the interval

$$r_{\min} \leq r \leq r_{\max}.$$

As the particle moves, its radial distance oscillates between these two values while its angular position continues to change. Since the particle cannot escape to infinity, the motion is bounded. Simply observing the effective potential, we cannot say anything about the trajectory of the particle; therefore, we need to solve the equations of motion to determine the actual path of the particle. This is what we will do in Section 11 and we will see that the resulting orbit is an ellipse, with the center of force located at one of its foci.

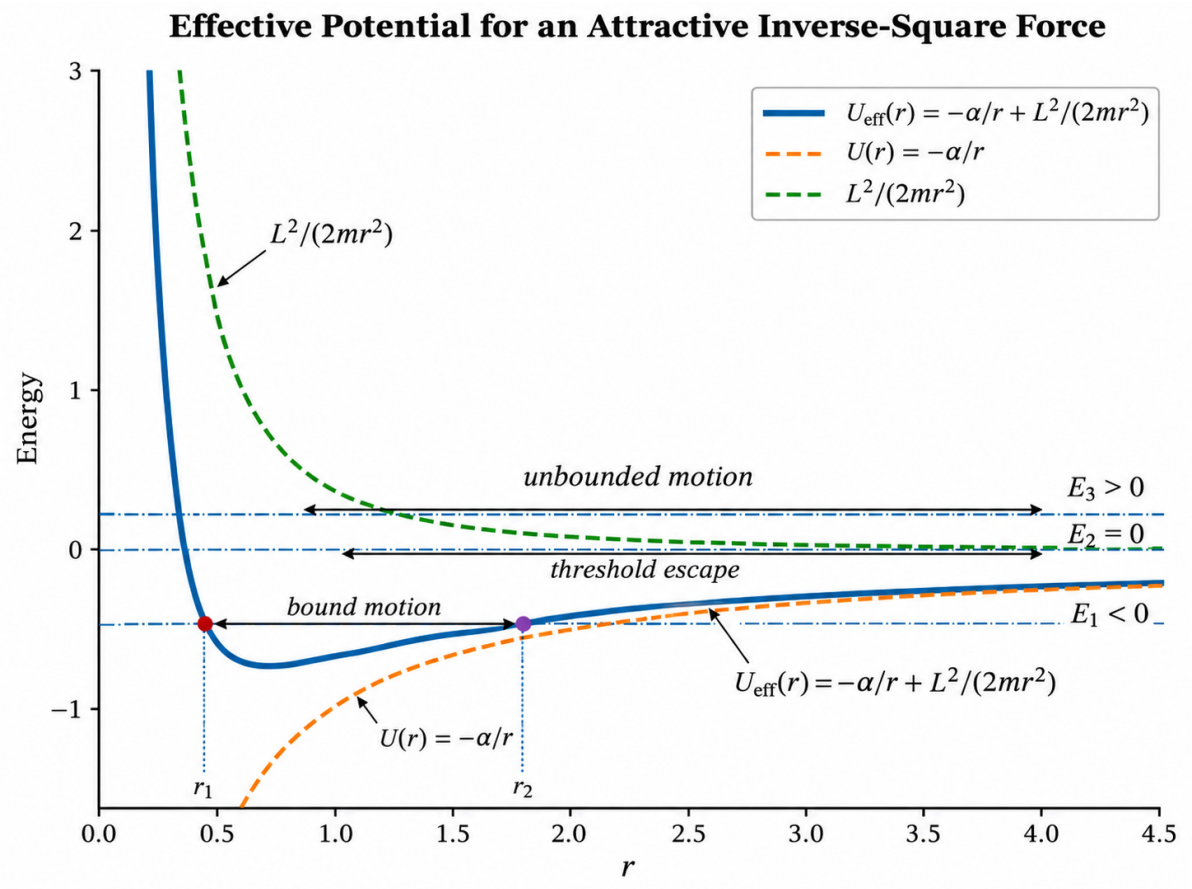


Figure 5: Effective potential for an attractive inverse-square force with nonzero angular momentum (solid curve), together with the attractive potential energy and the centrifugal potential energy (dashed curves). The horizontal lines represent different total energies. For $E_{\text{min}} < E_1 < 0$, the radial motion is confined between two turning points. For $E_2 = 0$, the particle is at the escape threshold, while for $E_3 > 0$, the motion is unbounded.

3. Zero energy ($E = 0$): When the total energy is exactly zero, the particle has just enough energy to escape to infinity. There is only one finite radial turning point, corresponding to the minimum distance from the center of force. Beyond this point, the particle can move to arbitrarily large distances. As $r \to \infty$, the effective potential approaches zero, so the radial velocity also approaches zero:

$$\dot{r} \to 0.$$

As we will see in Section 11, the resulting orbit is a parabola. This case represents the threshold between bounded and unbounded motion.

4. Positive energy ($E > 0$): For positive total energy, the particle is no longer gravitationally or electrostatically bound to the center of force. The energy line intersects the effective potential at only one positive radius, corresponding to the distance of closest approach. The particle can approach the center up to this distance and then move away toward infinity. At large distances, as $r \to \infty$, the effective potential vanishes, and conservation of energy gives

$$\frac{1}{2}m\dot{r}^2 \to E,$$

so that

$$|\dot{r}| \to \sqrt{\frac{2E}{m}}.$$

As we will see in Section 11, the resulting orbit is a hyperbola, and the particle escapes to infinity with a nonzero asymptotic speed.

Thus, the effective potential provides a simple way to classify the possible trajectories under an attractive inverse-square force: circular or elliptical orbits for negative total energy, parabolic orbits for zero energy, and hyperbolic orbits for positive total energy.

10 The Two-Body Problem

In the preceding sections, we studied the motion of a particle of mass m under a central force directed toward a fixed point. However, in many physical situations, such as the gravitational interaction between the Earth and the Sun, both interacting bodies are free to move. The Sun attracts the Earth, but the Earth also attracts the Sun with an equal and opposite force. Consequently, neither body is strictly stationary.

This raises an important question: *Can the motion of two interacting bodies be reduced to the central-force motion of a single particle?*

As we shall see, the answer is yes. By introducing the *center of mass* and the *relative position*, the two-body problem can be separated into two independent problems: the uniform motion of the center of mass and the motion of a fictitious particle under a central force.

10.1 Equations of Motion

Consider two particles with masses m_1 and m_2, whose position vectors in an inertial reference frame are $\mathbf{r}_1$ and $\mathbf{r}_2$, respectively. Suppose that the particles interact through a central force that depends only on their separation. We assume that there are no external forces acting on the system.We define the relative position vector as

$$\mathbf{r} = \mathbf{r}_1 - \mathbf{r}_2, \tag{14}$$

Thus, $\mathbf{r}$ points from particle 2 toward particle 1. Let $\mathbf{F}_{12}$ denote the force exerted on particle 1 by particle 2. Since the interaction is central,

$$\mathbf{F}_{12} = f(r)\mathbf{e}_r, \qquad \mathbf{e}_r = \frac{\mathbf{r}}{r}, \tag{15}$$

where $f(r) < 0$ corresponds to an attractive force. According to Newton's third law, the force exerted on particle 2 by particle 1 is

$$\mathbf{F}_{21} = -\mathbf{F}_{12}.$$

Newton's second law therefore gives

$$m_1\ddot{\mathbf{r}}_1 = f(r)\mathbf{e}_r, \tag{16}$$
$$m_2\ddot{\mathbf{r}}_2 = -f(r)\mathbf{e}_r. \tag{17}$$

Although these equations are coupled, we can transform them into two independent equations by introducing suitable coordinates.

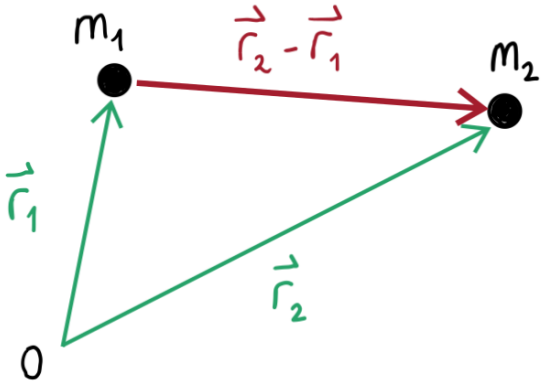


Figure 6: Two particles interacting through a central force. The relative position vector $\mathbf{r}$ points from particle 2 toward particle 1, while $\mathbf{R}$ denotes the center-of-mass position. The interaction forces are equal and opposite.

10.2 Motion of the Center of Mass

The center of mass of the two particles is defined by

$$\mathbf{R} = \frac{m_1\mathbf{r}_1 + m_2\mathbf{r}_2}{m_1 + m_2}. \tag{18}$$

Let us introduce the total mass

$$M = m_1 + m_2.$$

Differentiating Eq. (18) twice with respect to time gives

$$M\ddot{\mathbf{R}} = m_1\ddot{\mathbf{r}}_1 + m_2\ddot{\mathbf{r}}_2.$$

Using Newton's equations, we obtain

$$M\ddot{\mathbf{R}} = \mathbf{F}_{12} + \mathbf{F}_{21} = 0.$$

Hence,

$$\boxed{\ddot{\mathbf{R}} = 0.} \tag{19}$$

Therefore, the center of mass moves with constant velocity:

$$\mathbf{R}(t) = \mathbf{R}_0 + \mathbf{V}_{\mathrm{CM}}t. \tag{20}$$

This is a direct consequence of conservation of total linear momentum. Indeed,

$$\mathbf{P} = m_1\dot{\mathbf{r}}_1 + m_2\dot{\mathbf{r}}_2 = M\dot{\mathbf{R}}$$

is constant.

Since the center of mass moves uniformly, we may choose an inertial reference frame moving with it. In this *center-of-mass frame*,

$$\mathbf{R} = 0, \qquad \dot{\mathbf{R}} = 0.$$

The remaining problem is then to determine how the two particles move relative to one another.

10.3 Relative Motion and the Reduced Mass

We now subtract the equations of motion of the two particles.

From Eqs. (16) and (17), we have

$$\ddot{\mathbf{r}}_1 = \frac{\mathbf{F}_{12}}{m_1}, \qquad \ddot{\mathbf{r}}_2 = -\frac{\mathbf{F}_{12}}{m_2}.$$

Since

$$\ddot{\mathbf{r}} = \ddot{\mathbf{r}}_1 - \ddot{\mathbf{r}}_2,$$

we obtain

$$\ddot{\mathbf{r}} = \left(\frac{1}{m_1} + \frac{1}{m_2}\right) \mathbf{F}_{12}.$$

We introduce the *reduced mass*

$$\boxed{\mu = \frac{m_1 m_2}{m_1 + m_2}.} \tag{21}$$

Notice that

$$\frac{1}{\mu} = \frac{1}{m_1} + \frac{1}{m_2}.$$

The equation for the relative motion therefore becomes

$$\boxed{\mu \ddot{\mathbf{r}} = f(r)\mathbf{e}_r.} \tag{22}$$

This equation has exactly the same form as Newton's second law for a single particle of mass μ moving under a central force.

Thus, the original two-body problem is mathematically equivalent to the motion of a single fictitious particle with mass μ and position vector $\mathbf{r}$, subjected to the same central-force law.

Physical interpretation of the reduced mass.

The reduced mass takes into account the fact that both particles accelerate under their mutual interaction.

For example, when the particles move toward one another, their relative acceleration is the sum of the magnitudes of their individual accelerations. The reduced mass is the mass that reproduces this relative acceleration when the interaction force acts on a single fictitious particle.

It is important to emphasize that this fictitious particle describes the *relative motion*, not the actual motion of either individual particle.

Two limiting cases are particularly useful:

- If $m_2 \gg m_1$, then

$$\mu = \frac{m_1}{1 + m_1/m_2} \approx m_1.$$

  The heavier body moves very little, and the lighter body can approximately be treated as a particle orbiting a fixed center.

- If $m_1 = m_2 = m$, then
$$\mu = \frac{m}{2}.$$
Both particles contribute equally to the relative acceleration.

10.4 Recovering the Motion of Both Particles

After solving the equation for the relative position $\mathbf{r}(t)$, how do we determine the actual positions of the two particles?

We have the two definitions

$$\mathbf{r} = \mathbf{r}_1 - \mathbf{r}_2,$$
$$M\mathbf{R} = m_1\mathbf{r}_1 + m_2\mathbf{r}_2.$$

Solving these equations for $\mathbf{r}_1$ and $\mathbf{r}_2$ gives

$$\boxed{\mathbf{r}_1 = \mathbf{R} + \frac{m_2}{M}\mathbf{r},} \tag{23}$$

$$\boxed{\mathbf{r}_2 = \mathbf{R} - \frac{m_1}{M}\mathbf{r}.} \tag{24}$$

Therefore, once $\mathbf{R}(t)$ and $\mathbf{r}(t)$ are known, the positions of both particles are completely determined.

In the center-of-mass frame, where $\mathbf{R} = 0$, these expressions simplify to

$$\mathbf{r}_1 = \frac{m_2}{M}\mathbf{r}, \tag{25}$$
$$\mathbf{r}_2 = -\frac{m_1}{M}\mathbf{r}. \tag{26}$$

The particles therefore always lie on opposite sides of their common center of mass, with distances

$$r_1 = \frac{m_2}{M}r, \qquad r_2 = \frac{m_1}{M}r. \tag{27}$$

Consequently,

$$\boxed{\frac{r_1}{r_2} = \frac{m_2}{m_1}.} \tag{28}$$

The more massive particle always remains closer to the center of mass.

The circular trajectories shown in Fig. 7 represent a special case. For other allowed motions, the particles follow different trajectories. Nevertheless, their position vectors relative to the center of mass remain proportional to the same vector $\mathbf{r}(t)$.

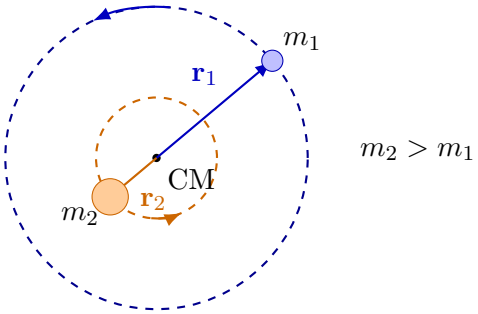


Figure 7: Two particles executing circular motion about their common center of mass. The more massive particle follows the smaller circle. The particles remain on opposite sides of the center of mass and have the same angular velocity. The dashed circles represent their trajectories in the center-of-mass frame.

In particular, for an inverse-square attractive force, the relative trajectory can be an ellipse, parabola, or hyperbola, depending on the energy. The individual trajectories in the center-of-mass frame have the same geometrical shape as the relative trajectory, but are scaled according to Eqs. (23) and (24).

10.5 Conservation of Energy

Let us now investigate how the total mechanical energy of the two-body system is related to the energy of the fictitious particle.

The total kinetic energy is

$$T = \frac{1}{2}m_1\dot{\mathbf{r}}_1^2 + \frac{1}{2}m_2\dot{\mathbf{r}}_2^2. \tag{29}$$

Differentiating Eqs. (23) and (24), we obtain

$$\dot{\mathbf{r}}_1 = \dot{\mathbf{R}} + \frac{m_2}{M}\dot{\mathbf{r}},$$
$$\dot{\mathbf{r}}_2 = \dot{\mathbf{R}} - \frac{m_1}{M}\dot{\mathbf{r}}.$$

Substituting these expressions into the kinetic energy and expanding the squares gives

$$T = \frac{1}{2}m_1\left(\dot{\mathbf{R}}^2 + 2\frac{m_2}{M}\dot{\mathbf{R}}\cdot\dot{\mathbf{r}} + \frac{m_2^2}{M^2}\dot{\mathbf{r}}^2\right) + \frac{1}{2}m_2\left(\dot{\mathbf{R}}^2 - 2\frac{m_1}{M}\dot{\mathbf{R}}\cdot\dot{\mathbf{r}} + \frac{m_1^2}{M^2}\dot{\mathbf{r}}^2\right).$$

Notice that the terms involving $\dot{\mathbf{R}}\cdot\dot{\mathbf{r}}$ cancel. Collecting the remaining terms, we find

$$\boxed{T = \frac{1}{2}M\dot{\mathbf{R}}^2 + \frac{1}{2}\mu\dot{\mathbf{r}}^2.} \tag{30}$$

Thus, the kinetic energy separates into two parts: the kinetic energy of the center of mass and the kinetic energy associated with relative motion.

For a conservative central interaction, the potential energy depends only on the separation r:

$$U = U(r), \qquad f(r) = -\frac{dU}{dr}.$$

The total mechanical energy is therefore

$$E_{\text{tot}} = \frac{1}{2}M\dot{\mathbf{R}}^2 + \frac{1}{2}\mu\dot{\mathbf{r}}^2 + U(r). \tag{31}$$

Since $\dot{\mathbf{R}}$ is constant, the energy associated with the relative motion is separately conserved:

$$\boxed{E = \frac{1}{2}\mu\dot{\mathbf{r}}^2 + U(r) = \text{constant}.} \tag{32}$$

Here, E denotes the energy of the relative motion. It equals the total mechanical energy in the center-of-mass frame.

10.6 Angular Momentum and the Effective Potential

Because the force acting on the fictitious particle is central, its torque about the origin of the relative coordinate system vanishes:

$$\boldsymbol{\tau} = \mathbf{r} \times \mathbf{F}_{12} = 0.$$

Consequently, the angular momentum associated with relative motion,

$$\boxed{\mathbf{L} = \mu\mathbf{r} \times \dot{\mathbf{r}},} \tag{33}$$

is conserved.

The vector $\mathbf{r}$ therefore moves in a fixed plane perpendicular to $\mathbf{L}$, provided $\mathbf{L} \neq 0$. We can describe the relative motion using polar coordinates (r, θ) in this plane.

As derived previously for central-force motion,

$$L = \mu r^2\dot{\theta}.$$

The relative kinetic energy can be written as

$$T_{\text{rel}} = \frac{1}{2}\mu\dot{r}^2 + \frac{1}{2}\mu r^2\dot{\theta}^2.$$

Using the conservation of angular momentum, we obtain

$$\frac{1}{2}\mu r^2\dot{\theta}^2 = \frac{L^2}{2\mu r^2}.$$

The conservation of energy then takes the form

$$\boxed{E = \frac{1}{2}\mu\dot{r}^2 + U_{\text{eff}}(r),} \tag{34}$$

where

$$\boxed{U_{\text{eff}}(r) = U(r) + \frac{L^2}{2\mu r^2}.} \tag{35}$$

This is exactly the effective-potential formalism developed in the preceding sections, with the ordinary particle mass replaced by the reduced mass μ.

The radial equation of motion is correspondingly

$$\mu\ddot{r} = -\frac{dU_{\text{eff}}}{dr} = f(r) + \frac{L^2}{\mu r^3}. \tag{36}$$

We can therefore use all the results obtained for central-force motion to analyze the relative motion of two interacting particles.

In particular, the turning points of the separation between the particles satisfy

$$E = U_{\text{eff}}(r),$$

while circular relative motion occurs at a stationary point of the effective potential:

$$\frac{dU_{\text{eff}}}{dr} = 0.$$

One important distinction should be kept in mind: the radial coordinate r represents the *distance between the two particles*, not the distance of either particle from the center of mass.

10.7 Example: The Gravitational Two-Body Problem

Consider two bodies with masses m_1 and m_2 interacting through Newton's gravitational force.

The force acting on particle 1 is

$$\mathbf{F}_{12} = -\frac{Gm_1m_2}{r^2}\mathbf{e}_r, \tag{37}$$

where G is the gravitational constant.

The relative equation of motion is therefore

$$\mu\ddot{\mathbf{r}} = -\frac{Gm_1m_2}{r^2}\mathbf{e}_r. \tag{38}$$

Dividing by the reduced mass gives

$$\boxed{\ddot{\mathbf{r}} = -\frac{G(m_1+m_2)}{r^2}\mathbf{e}_r.} \tag{39}$$

Notice an interesting result: the relative acceleration depends on the *sum of the two masses*.

The gravitational potential energy is

$$U(r) = -\frac{Gm_1m_2}{r},$$

and the effective potential becomes

$$U_{\text{eff}}(r) = -\frac{Gm_1m_2}{r} + \frac{L^2}{2\mu r^2}. \tag{40}$$

This has precisely the same mathematical form as the inverse-square effective potential studied in the previous section.

For a circular orbit, the separation r is constant, so the relative radial acceleration must satisfy

$$\mu r\omega^2 = \frac{Gm_1m_2}{r^2}.$$

Using the definition of the reduced mass, we find

$$\boxed{\omega^2 = \frac{G(m_1+m_2)}{r^3}.} \tag{41}$$

Since the orbital period is $T = 2\pi/\omega$,

$$\boxed{T^2 = \frac{4\pi^2}{G(m_1+m_2)}r^3.} \tag{42}$$

This is the circular-orbit form of Kepler's third law for a genuine two-body system.

The two bodies move along circles with radii

$$r_1 = \frac{m_2}{m_1+m_2}r, \qquad r_2 = \frac{m_1}{m_1+m_2}r,$$

and both circles are traversed with the same angular velocity ω.

For a system such as the Sun and the Earth, $m_{\text{Sun}} \gg m_{\text{Earth}}$. Consequently, the center of mass lies very close to the center of the Sun, and the reduced mass is approximately equal to the mass of the Earth.

This explains why treating the Sun as a fixed center of force is an excellent approximation in elementary orbital mechanics. Nevertheless, the exact two-body description shows that both bodies move around their common center of mass.

10.8 Summary

The essential result of the two-body problem is that the motion of two particles interacting through a central force can be separated into:

1. The uniform motion of their center of mass,
$$M\ddot{\mathbf{R}} = 0;$$
2. The relative motion of a fictitious particle with reduced mass μ,
$$\mu\ddot{\mathbf{r}} = f(r)\mathbf{e}_r.$$

The original particle positions can then be recovered from the center-of-mass and relative coordinates:
$$\mathbf{r}_1 = \mathbf{R} + \frac{m_2}{M}\mathbf{r}, \qquad \mathbf{r}_2 = \mathbf{R} - \frac{m_1}{M}\mathbf{r}.$$

Therefore, the central-force problem studied so far is not restricted to a particle moving around a physically fixed center. It also provides an exact description of the relative motion of two interacting particles, provided their interaction is central and there are no external forces.

In the next section, we will solve the equations of motion for the inverse-square force and determine the possible orbital trajectories.

11 Solution of the Equations of Motion

12 Worked Examples

12.1 Attractive Inverse-Square Force

A particle moves in the potential
$$U(r) = -\frac{\alpha}{r}, \qquad \alpha > 0.$$

For a fixed nonzero angular momentum L, determine the circular orbit radius and classify the radial motion according to the total energy.

Solution:

The effective potential is

$$U_{\text{eff}}(r) = -\frac{\alpha}{r} + \frac{L^2}{2mr^2}.$$

It tends to $+\infty$ as $r \to 0$ and approaches zero from below as $r \to \infty$.

Its stationary point satisfies

$$\frac{dU_{\text{eff}}}{dr} = \frac{\alpha}{r^2} - \frac{L^2}{mr^3} = 0,$$

giving

$$r_0 = \frac{L^2}{m\alpha}.$$

At this radius,

$$U''_{\text{eff}}(r_0) = \frac{\alpha}{r_0^3} > 0,$$

so the circular orbit is stable. Its energy is

$$E_{\text{c}} = U_{\text{eff}}(r_0) = -\frac{m\alpha^2}{2L^2}.$$

The turning-point equation is

$$E = -\frac{\alpha}{r} + \frac{L^2}{2mr^2},$$

or

$$Er^2 + \alpha r - \frac{L^2}{2m} = 0.$$

For this particular potential, the equation is quadratic in r when $E \neq 0$.

The possible motions are:

- $E < E_{\text{c}}$: no motion is possible for the specified angular momentum.
- $E = E_{\text{c}}$: the orbit is circular, with $r = r_0$.
- $E_{\text{c}} < E < 0$: there are two positive turning points, and the motion is bounded.
- $E \geq 0$: there is one positive turning point, corresponding to the minimum radius, and the allowed region extends to infinity.

For $E = 0$, the minimum radius is

$$r_{\min} = \frac{L^2}{2m\alpha}.$$

This is the threshold between bounded and unbounded motion: the speed tends to zero as the particle moves to infinity.

12.2 Isotropic Harmonic Force

A particle moves under the central force

$$\mathbf{F} = -kr\,\mathbf{e}_r, \qquad k > 0.$$

For a fixed nonzero angular momentum L, find the turning radii and determine whether the motion can be unbounded.

Solution:

Choosing the potential energy to vanish at the origin gives

$$U(r) = \frac{1}{2}kr^2.$$

Thus,

$$U_{\text{eff}}(r) = \frac{1}{2}kr^2 + \frac{L^2}{2mr^2}.$$

The effective potential tends to $+\infty$ both as $r \to 0$ and as $r \to \infty$. Consequently, every allowed motion with finite energy is bounded.

The turning-point equation is

$$E = \frac{1}{2}kr^2 + \frac{L^2}{2mr^2},$$

or

$$kr^4 - 2Er^2 + \frac{L^2}{m} = 0.$$

This equation is quartic in r, but quadratic in r^2. Solving for r^2 gives

$$r^2_{\text{min}} = \frac{E - \sqrt{E^2 - kL^2/m}}{k}, \qquad r^2_{\text{max}} = \frac{E + \sqrt{E^2 - kL^2/m}}{k}.$$

Allowed motion requires

$$E \geq |L|\sqrt{\frac{k}{m}}.$$

At equality, the two radii coincide and the orbit is circular:

$$r_0 = \left(\frac{L^2}{mk}\right)^{1/4}.$$

Above this energy, the radius oscillates between two distinct turning points.

Unlike the inverse-square potential, the harmonic potential does not allow escape to infinity at any finite energy. Thus, the sign of the total energy alone is not a universal criterion for bounded motion; the shape and chosen zero of the potential must be taken into account.

12.3 Attractive and Repulsive Forces

Decide if the following forces are attractive or repulsive for $k > 0$.

(a) $\mathbf{F} = -\frac{3k}{\sqrt{r}}\mathbf{e}_r$

(b) $\mathbf{F} = 4kr^2\mathbf{e}_r$

(c) $\mathbf{F} = r(r-2)\mathbf{e}_r$

Solution:

(a) The force is attractive because the negative sign indicates that the force is directed towards the origin, and the factor $\frac{3k}{\sqrt{r}}$ is positive for $k > 0$.

(b) The force is repulsive because the positive sign indicates that the force is directed away from the origin, and the factor $4kr^2$ is positive for $k > 0$.

(c) The force is attractive for $0 < r < 2$ and repulsive for $r > 2$. At $r = 0$, the force is undefined, and at $r = 2$, the force is zero.

12.4

If a particle moves in a plane according to expressions

$$\theta(t) = 0.3t + 0.2t^2, \quad r(t) = 0.5t + 0.4t^2$$

Find its velocity and acceleration at $t = 2$ s.

Solution:

12.5 Finding force-law from trajectory

Find the force law for the central force that allows a particle to move in a logaritmic spiral trajectory given by $r = r_0 e^{k\theta}$, where r_0 and k are constants. Determine $r(t)$ and $\theta(t)$ for the particle.

Solution:

12.6 Determining the Central Force from an Elliptical Orbit

A particle of mass m moves along an elliptical orbit under the influence of a central force directed toward one of the foci of the ellipse. The trajectory is described in polar coordinates

by

$$r(\theta) = \frac{p}{1 + e\cos\theta}, \tag{43}$$

where $p > 0$ is a constant and $0 < e < 1$ is the eccentricity of the ellipse.

Using Binet's equation, determine the central force law $F(r)$ that produces this orbit. Express your answer in terms of the particle's angular momentum L, mass m, and the parameter p.

Solution:

Binet's equation is

$$F(r) = -\frac{L^2}{m}u^2\left(\frac{d^2u}{d\theta^2} + u\right), \qquad u = \frac{1}{r}. \tag{44}$$

From the given trajectory,

$$u(\theta) = \frac{1 + e\cos\theta}{p}. \tag{45}$$

Differentiating twice with respect to θ,

$$\frac{du}{d\theta} = -\frac{e}{p}\sin\theta, \qquad \frac{d^2u}{d\theta^2} = -\frac{e}{p}\cos\theta. \tag{46}$$

Therefore,

$$\frac{d^2u}{d\theta^2} + u = -\frac{e}{p}\cos\theta + \frac{1 + e\cos\theta}{p} = \frac{1}{p}. \tag{47}$$

Substituting into Binet's equation gives

$$F(r) = -\frac{L^2}{mp}u^2. \tag{48}$$

Since $u = 1/r$, we obtain

$$\boxed{F(r) = -\frac{L^2}{mp}\frac{1}{r^2}}. \tag{49}$$

Thus, an elliptical orbit with the center of force at one of its foci is produced by an attractive inverse-square force.

12.7 Work and kinetic energy in a rotating system

Two identical small masses, each of mass m, are connected to a frictionless vertical rotation axis by two massless rods of adjustable length. The system rotates in a horizontal plane with an initial angular velocity ω_i, and each mass is initially at a distance r_i from the rotation axis. The rods are then shortened, pulling the masses radially inward until their distance from the axis becomes $r_f = r_i/\sqrt{2}$. Assume that no external torque acts about the rotation axis and that the radial velocities of the masses are zero both initially and finally.

(a) Calculate the change in the total kinetic energy of the system using conservation of angular momentum.

(b) Calculate the work done by the pulling forces directly from the definition of work, $W = \int \mathbf{F} \cdot d\mathbf{r}$, and verify that it agrees with the result obtained in part (a).

Solution:

(a) Since no external torque acts about the rotation axis, the total angular momentum of the system is conserved. Treating the two masses as point particles, the initial moment of inertia is

$$I_i = 2mr_i^2. \tag{50}$$

After the masses are pulled inward to $r_f = r_i/\sqrt{2}$, the final moment of inertia becomes

$$I_f = 2mr_f^2 = 2m\left(\frac{r_i}{\sqrt{2}}\right)^2 = mr_i^2 = \frac{I_i}{2}. \tag{51}$$

From conservation of angular momentum, we have

$$I_i\omega_i = I_f\omega_f \implies 2mr_i^2\omega_i = mr_i^2\omega_f. \tag{52}$$

Therefore, the final angular velocity is

$$\boxed{\omega_f = 2\omega_i}. \tag{53}$$

Since the radial velocities are zero initially and finally, the kinetic energy at these instants is purely rotational. The initial kinetic energy is

$$K_i = \frac{1}{2}I_i\omega_i^2 = mr_i^2\omega_i^2. \tag{54}$$

Similarly, the final kinetic energy is

$$K_f = \frac{1}{2}I_f\omega_f^2 = \frac{1}{2}(mr_i^2)(2\omega_i)^2 = 2mr_i^2\omega_i^2. \tag{55}$$

Thus, the change in kinetic energy is

$$\Delta K = K_f - K_i = 2mr_i^2\omega_i^2 - mr_i^2\omega_i^2, \tag{56}$$

which gives

$$\boxed{\Delta K = mr_i^2\omega_i^2}. \tag{57}$$

The kinetic energy of the system increases even though its angular momentum remains constant. This increase must come from the work performed by the forces pulling the masses inward.

(b) To calculate this work directly, we start from the definition of work,

$$W = \int \mathbf{F} \cdot d\mathbf{r}. \tag{58}$$

Since the pulling force acts in the radial direction, only the radial component of displacement contributes to the work. For each mass, we can therefore write

$$W_1 = \int_{r_i}^{r_f} F_r \, dr, \tag{59}$$

where F_r is the radial component of the pulling force, with the outward radial direction chosen as positive.

From Newton's second law in polar coordinates, the radial acceleration of a particle is

$$a_r = \ddot{r} - r\omega^2. \tag{60}$$

Therefore, the radial force acting on each mass is

$$F_r = m(\ddot{r} - r\omega^2). \tag{61}$$

Since there are two identical masses, the total work done by the pulling forces is

$$W = 2\int_{r_i}^{r_f} F_r \, dr = 2m\int_{r_i}^{r_f} \ddot{r} \, dr - 2m\int_{r_i}^{r_f} r\omega^2 \, dr. \tag{62}$$

Let us first evaluate the integral containing the radial acceleration. Using

$$\ddot{r} = \frac{d\dot{r}}{dt}, \qquad dr = \dot{r} \, dt, \tag{63}$$

we obtain

$$\int_{r_i}^{r_f} \ddot{r} \, dr = \int_{\dot{r}_i}^{\dot{r}_f} \dot{r} \, d\dot{r} = \frac{1}{2}\left(\dot{r}_f^2 - \dot{r}_i^2\right). \tag{64}$$

Since the radial velocities are zero both initially and finally, this integral vanishes. Hence, the total work reduces to

$$W = -2m\int_{r_i}^{r_f} r\omega^2 \, dr. \tag{65}$$

However, the angular velocity is not constant during the inward motion. To express it as a function of the radial position r, we use conservation of angular momentum:

$$I(r)\omega(r) = I_i\omega_i. \tag{66}$$

Since $I(r) = 2mr^2$, we have

$$2mr^2\omega(r) = 2mr_i^2\omega_i, \tag{67}$$

which gives

$$\omega(r) = \frac{r_i^2}{r^2}\omega_i. \tag{68}$$

Substituting this expression into the work integral, we obtain

$$W = -2m \int_{r_i}^{r_f} r \left(\frac{r_i^2}{r^2}\omega_i\right)^2 dr \tag{69}$$

$$= -2mr_i^4\omega_i^2 \int_{r_i}^{r_f} \frac{dr}{r^3}. \tag{70}$$

Evaluating the integral,

$$W = -2mr_i^4\omega_i^2 \left[-\frac{1}{2r^2}\right]_{r_i}^{r_f} \tag{71}$$

$$= mr_i^4\omega_i^2 \left(\frac{1}{r_f^2} - \frac{1}{r_i^2}\right). \tag{72}$$

Finally, substituting $r_f = r_i/\sqrt{2}$, we find

$$W = mr_i^4\omega_i^2 \left(\frac{2}{r_i^2} - \frac{1}{r_i^2}\right) \tag{73}$$

$$= mr_i^2\omega_i^2. \tag{74}$$

Therefore,

$$\boxed{W = mr_i^2\omega_i^2}. \tag{75}$$

This result agrees exactly with the change in kinetic energy calculated in part (a):

$$\boxed{W = \Delta K = K_f - K_i}. \tag{76}$$

Thus, although the angular momentum of the system is conserved, its kinetic energy increases due to the positive work performed by the pulling forces. This demonstrates that conservation of angular momentum does not necessarily imply conservation of kinetic energy.